

PYTHON PROGRAMMING INDEX

320 PYTHON PRACTICE PROGRAMS

BEGINNER → ADVANCED

FIRST-YEAR B.TECH ROADMAP

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SECTION 1: BASIC PROGRAMS (1-20)

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1. Hello World
2. Print Name
3. Print College Details
4. Print Multiple Lines
5. Print ASCII Values
6. Add Two Numbers
7. Subtract Two Numbers
8. Multiply Two Numbers
9. Divide Two Numbers
10. Find Remainder
11. Swap Two Numbers (Using Third Variable)
12. Swap Without Third Variable
13. Square of a Number
14. Cube of a Number
15. Area of Circle
16. Area of Rectangle

- 17. Area of Triangle
- 18. Simple Interest
- 19. Compound Interest
- 20. Temperature Converter

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SECTION 2: DECISION MAKING (21-40)

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- 21. Even or Odd
- 22. Positive, Negative or Zero
- 23. Largest of Two Numbers
- 24. Largest of Three Numbers
- 25. Smallest of Three Numbers
- 26. Leap Year Check
- 27. Vowel or Consonant
- 28. Alphabet, Digit or Special Character
- 29. Uppercase or Lowercase
- 30. Grade Calculator
- 31. Percentage Calculator
- 32. Electricity Bill Calculator
- 33. Income Tax Calculator
- 34. BMI Calculator
- 35. Voting Eligibility

- 36. Age Category
- 37. Calculator Using if-else
- 38. Calculator Using match-case
- 39. Divisibility Check
- 40. Profit or Loss Calculator

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SECTION 3: LOOP PROGRAMS (41-80)

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- 41. Print 1 to N
- 42. Print N to 1
- 43. Sum of N Numbers
- 44. Sum of Even Numbers
- 45. Sum of Odd Numbers
- 46. Multiplication Table
- 47. Reverse Multiplication Table
- 48. Factorial
- 49. Fibonacci Series
- 50. Prime Number
- 51. Prime Numbers in Range
- 52. Reverse Number
- 53. Palindrome Number
- 54. Armstrong Number

- 55. Perfect Number
- 56. Strong Number
- 57. Neon Number
- 58. Harshad Number
- 59. Automorphic Number
- 60. Duck Number
- 61. Spy Number
- 62. Sunny Number
- 63. Peterson Number
- 64. Disarium Number
- 65. Happy Number
- 66. Krishnamurthy Number
- 67. GCD
- 68. LCM
- 69. Power Using Loop
- 70. Decimal to Binary
- 71. Binary to Decimal
- 72. Decimal to Octal
- 73. Decimal to Hexadecimal
- 74. Count Digits
- 75. Sum of Digits
- 76. Product of Digits
- 77. Largest Digit
- 78. Smallest Digit
- 79. Reverse Digits

80. Number Guessing Game

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SECTION 4: PATTERN PROGRAMS (81-120)

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STAR PATTERNS

81. Square

82. Rectangle

83. Right Triangle

84. Left Triangle

85. Inverted Triangle

86. Pyramid

87. Inverted Pyramid

88. Diamond

89. Hollow Square

90. Hollow Rectangle

91. Hollow Pyramid

92. Hollow Diamond

93. Butterfly

94. Sandglass

- 95. Hourglass
- 96. X Pattern
- 97. Plus Pattern
- 98. Cross Pattern
- 99. Arrow Pattern
- 100. Christmas Tree Pattern

NUMBER PATTERNS

- 101. Increasing Number Triangle
- 102. Decreasing Number Triangle
- 103. Floyd's Triangle
- 104. Pascal Triangle
- 105. Number Pyramid
- 106. Reverse Number Pyramid
- 107. Hollow Number Square
- 108. Number Diamond
- 109. Snake Number Pattern
- 110. Spiral Number Pattern

ALPHABET PATTERNS

- 111. Alphabet Triangle
- 112. Reverse Alphabet Triangle
- 113. Alphabet Pyramid
- 114. Alphabet Diamond
- 115. Alphabet Square
- 116. Hollow Alphabet Pattern
- 117. Character Pyramid
- 118. Alternate Alphabet Pattern
- 119. Alphabet Hourglass
- 120. Mixed Character Pattern

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SECTION 5: STRING PROGRAMS (121-150)

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- 121. Find String Length
- 122. Reverse String
- 123. Palindrome String
- 124. Count Vowels
- 125. Count Consonants
- 126. Count Digits
- 127. Count Spaces
- 128. Count Words

- 129. Count Characters
- 130. Character Frequency
- 131. Remove Spaces
- 132. Remove Duplicate Characters
- 133. Replace Characters
- 134. Toggle Case
- 135. Capitalize Sentence
- 136. Title Case
- 137. Check Anagram
- 138. Check Pangram
- 139. Find Longest Word
- 140. Find Shortest Word
- 141. Reverse Words
- 142. Find First Non-Repeating Character
- 143. Find Duplicate Characters
- 144. String Compression
- 145. Check Rotation
- 146. Count Special Characters
- 147. Extract Digits
- 148. Sort Characters
- 149. Compare Strings
- 150. Join Strings

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SECTION 6: LIST PROGRAMS (151-180)

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- 151. Create List
- 152. Sum of List
- 153. Average
- 154. Maximum Element
- 155. Minimum Element
- 156. Second Largest
- 157. Second Smallest
- 158. Reverse List
- 159. Sort List
- 160. Search Element
- 161. Count Frequency
- 162. Remove Duplicates
- 163. Merge Lists
- 164. Rotate List Left
- 165. Rotate List Right
- 166. Even Elements
- 167. Odd Elements
- 168. Positive Elements
- 169. Negative Elements
- 170. List Intersection
- 171. List Union
- 172. Common Elements

- 173. Missing Number
- 174. Pair Sum
- 175. Remove Element
- 176. Insert Element
- 177. Flatten Nested List
- 178. Matrix Using Lists
- 179. Transpose Matrix
- 180. Spiral Matrix

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SECTION 7: TUPLE PROGRAMS (181-190)

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- 181. Create Tuple
- 182. Convert List to Tuple
- 183. Tuple Packing
- 184. Tuple Unpacking
- 185. Nested Tuple
- 186. Count Elements
- 187. Maximum Element
- 188. Minimum Element
- 189. Tuple Slicing
- 190. Tuple Sorting

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SECTION 8: SET PROGRAMS (191-200)

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191. Union

192. Intersection

193. Difference

194. Symmetric Difference

195. Subset Check

196. Superset Check

197. Remove Duplicates

198. Frozen Set

199. Set Comprehension

200. Unique Elements

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SECTION 9: DICTIONARY PROGRAMS (201-220)

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201. Student Dictionary

202. Employee Dictionary

203. Word Counter

204. Character Frequency

- 205. Merge Dictionaries
- 206. Sort Dictionary
- 207. Maximum Value
- 208. Minimum Value
- 209. Dictionary Comprehension
- 210. Nested Dictionary
- 211. Phone Book
- 212. Inventory System
- 213. Student Marks
- 214. Attendance System
- 215. Shopping Cart
- 216. Contact Book
- 217. Library Records
- 218. Salary Database
- 219. Voting Counter
- 220. Cricket Scoreboard

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SECTION 10: FUNCTION PROGRAMS (221-240)

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- 221. Simple Function
- 222. Function with Arguments
- 223. Function Returning Value

- 224. Default Arguments
- 225. Keyword Arguments
- 226. Variable-Length Arguments
- 227. Recursive Factorial
- 228. Recursive Fibonacci
- 229. Recursive GCD
- 230. Recursive Binary Search
- 231. Lambda Calculator
- 232. Map Function
- 233. Filter Function
- 234. Reduce Function
- 235. Prime Using Function
- 236. Armstrong Using Function
- 237. Menu-Driven Functions
- 238. Function Decorator
- 239. Generator Function
- 240. Iterator Function

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SECTION 11: OOP PROGRAMS (241-260)

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- 241. Student Class
- 242. Employee Class

- 243. Bank Account
- 244. Rectangle Class
- 245. Circle Class
- 246. Car Class
- 247. Inheritance Example
- 248. Multiple Inheritance
- 249. Multilevel Inheritance
- 250. Hierarchical Inheritance
- 251. Method Overriding
- 252. Polymorphism
- 253. Abstract Class
- 254. Encapsulation
- 255. Operator Overloading
- 256. Magic Methods
- 257. Library Management
- 258. Hospital Management
- 259. ATM Class
- 260. Shopping Cart Class

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SECTION 12: FILE HANDLING (261-275)

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- 261. Create File

- 262. Read File
- 263. Write File
- 264. Append File
- 265. Count Characters
- 266. Count Words
- 267. Count Lines
- 268. Copy File
- 269. Merge Files
- 270. Search Word
- 271. CSV Reader
- 272. CSV Writer
- 273. JSON Reader
- 274. JSON Writer
- 275. Student Database File

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SECTION 13: EXCEPTION HANDLING (276-285)

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- 276. Divide by Zero
- 277. Invalid Input
- 278. File Not Found
- 279. Multiple Exceptions
- 280. Nested try-except

- 281. *finally* Block
- 282. *raise* Keyword
- 283. Custom Exception
- 284. ATM Exception
- 285. Login Validation

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SECTION 14: MODULES & LIBRARIES (286-295)

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- 286. Random Password Generator
- 287. Calendar Generator
- 288. Stopwatch
- 289. Digital Clock
- 290. QR Code Generator
- 291. URL Shortener
- 292. Weather App
- 293. Currency Converter
- 294. Calculator GUI
- 295. To-Do App

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SECTION 15: FINAL PROJECTS (296-320)

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- 296. Calculator
 - 297. Student Management System
 - 298. Library Management System
 - 299. Hospital Management System
 - 300. Banking System
 - 301. Inventory Management System
 - 302. Contact Book
 - 303. Expense Tracker
 - 304. Password Manager
 - 305. Quiz Application
 - 306. Attendance System
 - 307. Hotel Booking System
 - 308. Food Ordering System
 - 309. Online Shopping Cart
 - 310. Chat Application
 - 311. Portfolio Website (Flask)
 - 312. Blog Website (Django)
 - 313. Weather Dashboard
 - 314. AI Chatbot
 - 315. Voice Assistant
 - 316. Face Recognition Attendance
 - 317. Resume Generator
 - 318. File Organizer

319. URL Shortener Service

320. College Management System

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★ RECOMMENDED ORDER FOR FIRST-YEAR B.TECH

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|--------------------------------|--------------------|
| 1. Basics | → Programs 1-20 |
| 2. Decision Making | → Programs 21-40 |
| 3. Loops | → Programs 41-80 |
| 4. Patterns | → Programs 81-120 |
| 5. Strings | → Programs 121-150 |
| 6. Lists | → Programs 151-180 |
| 7. Tuples, Sets & Dictionaries | → Programs 181-220 |
| 8. Functions | → Programs 221-240 |
| 9. OOP | → Programs 241-260 |
| 10. File Handling | → Programs 261-275 |
| 11. Exception Handling | → Programs 276-285 |
| 12. Modules & Libraries | → Programs 286-295 |
| 13. Final Projects | → Programs 296-320 |

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BEGINNER



Basics



Decision Making



Loops



Patterns



Strings



Lists



Tuples



Sets



Dictionaries



Functions



OOP



File Handling

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Exception Handling

↓

Modules & Libraries

↓

Projects

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ADVANCED PYTHON PRACTICE

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 **FINAL TARGET**

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TOTAL PROGRAMS: 320

PYTHON GOAL:

- ✓ **Complete all 320 programs**
- ✓ **Understand the logic of every program**
- ✓ **Write programs without copying**
- ✓ **Practice debugging**
- ✓ **Maintain clean code**
- ✓ **Upload selected programs to GitHub**
- ✓ **Build projects after fundamentals**
- ✓ **Use Python for DSA and problem**